

AYMANE AIT DADS

Software Engineer Intern - Machine Learning & Data Fusion | Map Automation · Python · Intelligent Data-Driven Methods

Ayman.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the ongoing NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by building a full LoRA SFT training and evaluation pipeline on a 30B-parameter model - directly demonstrating the research, investigation, and automation skills HERE requires for HD map healing. Fine-tuned a CLIP ViT-L/14 (428M parameters) image-classification system to rank 1st in class and submitted to NTIRE 2026 at CVPR, achieving 0.791 AUC through ensemble methods and 10-view TTA on 250K training images. Brings production-grade Python, PyTorch, Scikit-learn, clustering pipelines, and ablation methodology to intelligent data-driven map fusion research. Aims to apply machine learning and automated evaluation techniques to HERE's High Definition Map quality problems, contributing rigorous testing and code that accelerates data fusion at scale.

CORE COMPETENCIES

Machine Learning | Data Fusion | Automation & Testing | Python Development | Data-Driven Methods | Evaluation Pipelines | Clustering & Classification | Research & Investigation

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built automated **machine learning pipeline** combining Random Forest classification and K-Means clustering on 100,000+ fiber-optic network records to evaluate network quality across thousands of nodes.
- Developed data-driven clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with outputs adopted directly by the network optimization team.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation scripts, and reproducible Python workflows using Pandas and Scikit-learn.
- Delivered stakeholder presentations translating **model outputs** into actionable network maintenance recommendations, bridging data science findings and operational decisions.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) across 5 structured reasoning categories.
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built diagnostic evaluation pipeline and custom cipher/bit-operation solvers revealing that reasoning over-expansion - not incorrect reasoning - was the primary failure mode.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

AI-Generated Image Detection - NTIRE 2026 @ CVPR

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class** (private Kaggle leaderboard, 0.791 AUC) and submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark using CLIP ViT-L/14 (428M parameters) fine-tuned on 250K images across 25+ generator types.
- Designed ensemble of 3 models weighted by validation AUC with **10-view TTA** (flips, crops, blur, resizes); discovered 1-epoch training outperformed 4-epoch training for out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Statistics, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML & Data-Driven Methods: PyTorch | Scikit-learn | LoRA SFT | Hugging Face Transformers | Clustering & Classification | Ensemble Methods | Ablation Studies

Programming & Automation: Python (expert) | SQL | Bash | Pandas | NumPy | Feature Engineering | Git

Computer Vision & Evaluation: CLIP ViT | CNN | DenseNet | FP16/BF16 Mixed Precision | Power BI | Matplotlib | Kaggle GPU